CSE 318 — Online 4

Section C1

Once the temperature has dropped, SA essentially becomes a pure hill-climber and gets trapped in the nearest local optimum. A common fix is *reheating*: if the best-known cost has not improved for a while, jump the temperature back up so the algorithm can escape and try a different region.

What to implement:
Modify your `simulatedAnnealing()` main loop as follows:

1. Keep track of `bestCost` (already done) and add a counter `stagnation` that counts the number of consecutive iterations in which `bestCost` did not improve.
2. Whenever a strictly better solution is found, reset `stagnation` to 0.
3. If `stagnation` reaches a threshold S (default $S = 5$), **reheat**: set $T \leftarrow T_0/2$, reset `stagnation` to 0, and increment a counter `reheatCount`.
4. Cap the number of reheats at $R = 3$ (after that, do no more reheats but continue until the iteration budget runs out).
5. Everything else (neighbor operator, cooling schedule, acceptance rule) stays identical.

Output

- Run modified SA **5 times** on a test case and report **best, average, worst** cost.
- Report the number of reheats that actually occurred in each run.